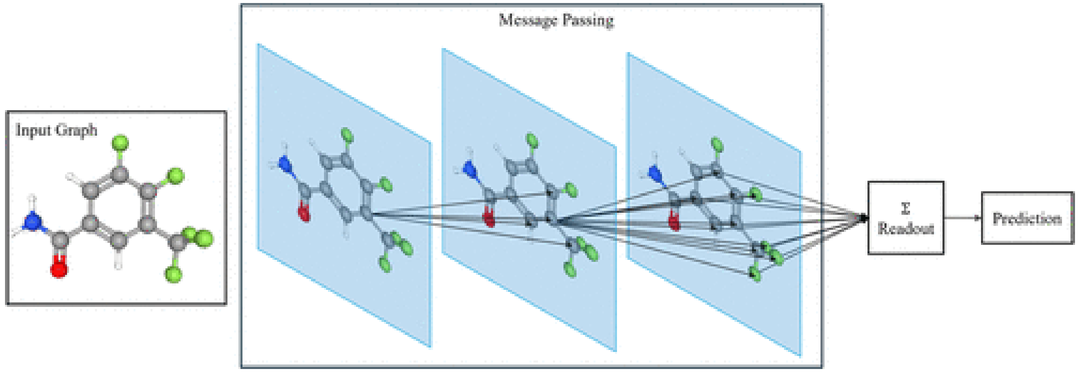
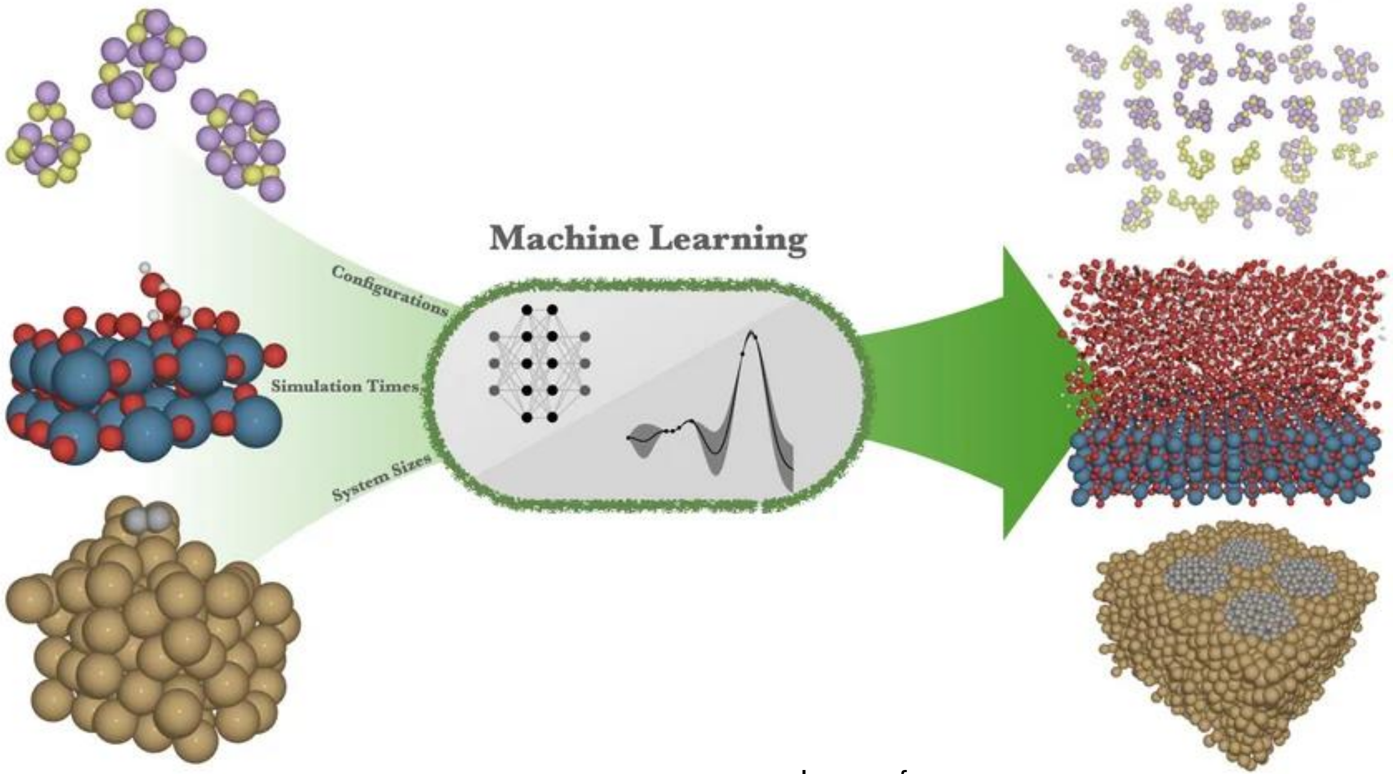
Machine Learning Interatomic Potentials Tutorial





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Images from:
https://www.fhi.mpg.de/739215/heenen_group
J. Chem. Inf. Model. 2026, 66, 3, 1316–1336

Goal of this tutorial

Understand what MLIP tools are applicable to your research and how to use each tool.

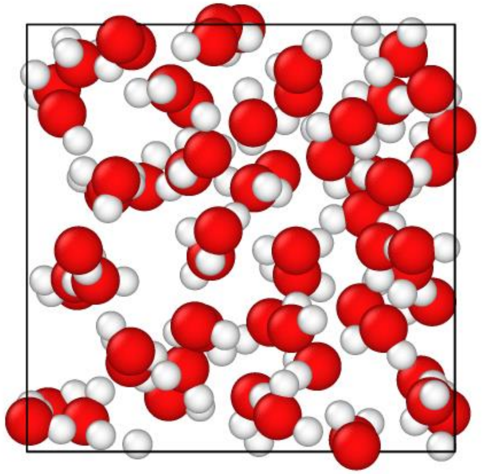
Agenda

1. MACE tutorial – github colab
   a. Exploring dataset
   b. Training a model
   c. Evaluating the model’s accuracy
   d. Understanding model parameters
   e. Try it out on Water!
2. UMA universal atomic simulator
   a. Visualizing molecular motion and computing energies/forces on the fly

1. Understanding the Data

Atomic structures (training data):

- Snapshots from molecular simulation
- Much smaller than the simulation size of interest (keeps the DFT cheap)



DFT computes energies and forces for each structure

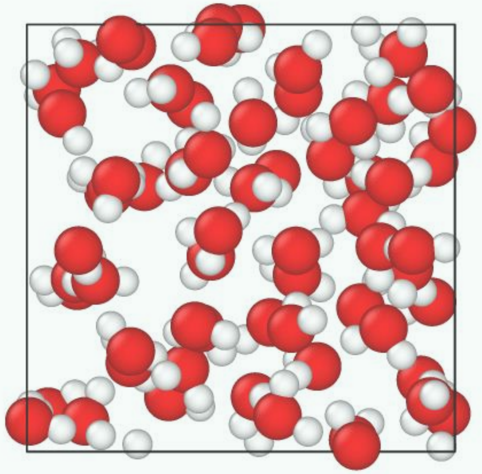
Training data:

$$\left\{\begin{matrix} \boldsymbol{Z} \\ E_{tot} \\ \boldsymbol{F} \end{matrix}\right\}$$

1. Understanding the Data

Key considerations:

- Do you sample the distribution of structures you want to model?
- Is the quality of theory in the DFT high enough for your question?



Must cover a large range of geometries

DFT computes energies and forces for each structure

@ a sufficient level of theory

Training data:

$$\begin{Bmatrix} \boldsymbol{Z} \\ E_{tot} \\ \boldsymbol{F} \end{Bmatrix}$$

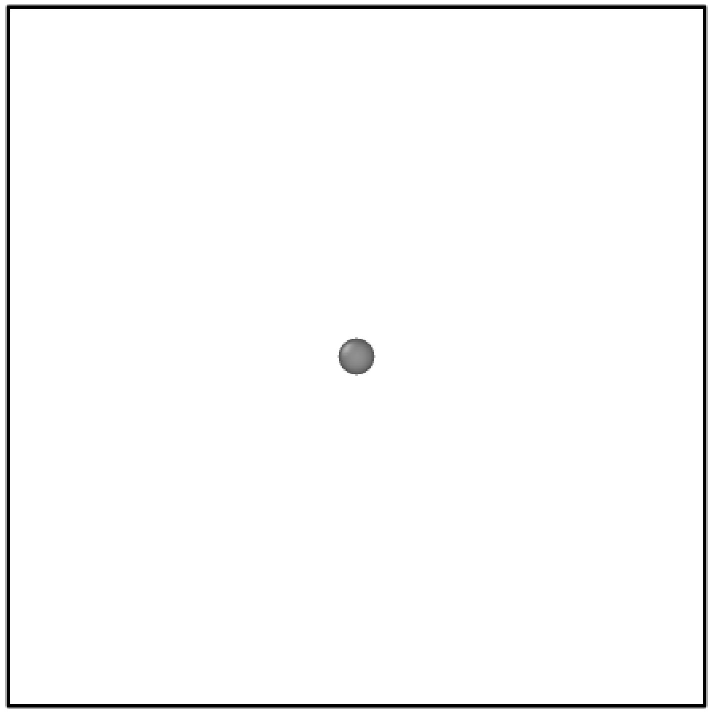
1. Understanding the Data

Isolated atom energies:

- Can either be given in the data
- Otherwise, the model learns them via regression

MLIPs fit to the atomization energy:

$$E^{atm} = E^{tot} - \sum_i E^0$$



E^0 can be computed by running DFT on the isolated atom (make sure to converge to box size)

2. Model Parameters

Hyper-parameters to change:

← **Speed**

64 128 256

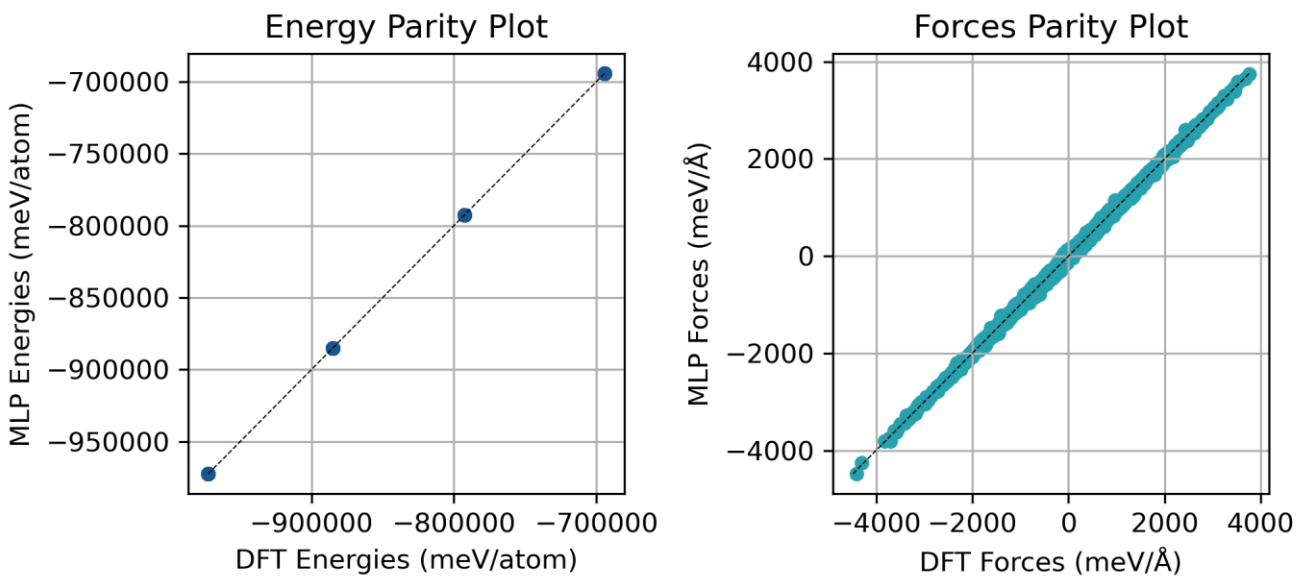
Model size →

- num_channels: size of the model
- max_L: symmetry of the messages
  - $L = 0$ trains on only scalar features
  - $L = 1$ trains on scalar and vector features (4 feature per channel)
- r_max: cutoff radius
  - Neighborhood on interaction between atoms
  - effective interaction cutoff is num_interactions $*$ r_max

3. Fitting and Testing MACE models

Testing models accuracy on reproducing DFT

- How well did the model train?
    - Test set errors
    - Parity plots



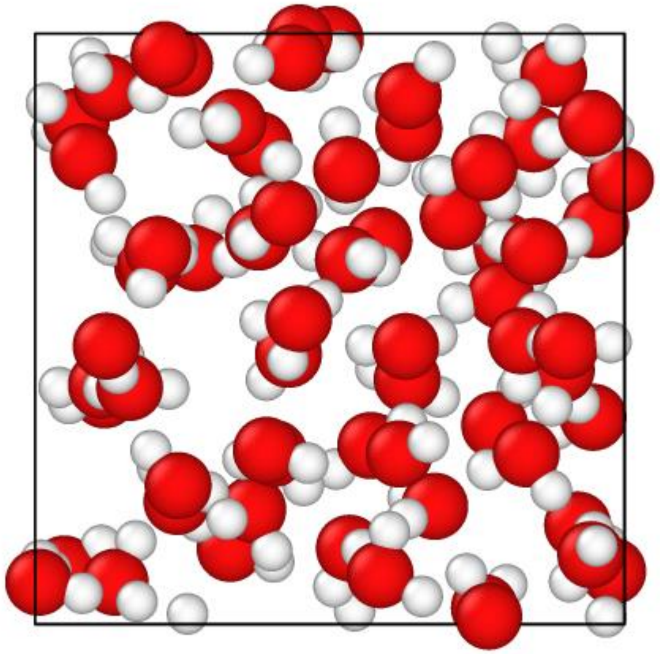
4. Running Molecular Dynamics

Testing model accuracy on prediction solution phase properties

Things to check:

- Stable dynamics
- Accurate dynamics – does it reproduce experimental quantities of interest?
  - Density
  - g(r) / structure factor
  - Transport properties (conductivity, self-diffusion)

Part B – Now try it for yourself on water!



UMA (Universal Model for Atoms) :

https://aidemos.atmeta.com/uma